for all $\sigma \geq 1$ and $|t| \geq 1$. Thus

$$(3) \quad |\zeta(\sigma + it)| \geq c'(\sigma - 1)^{3/4}|t|^{-\epsilon/4}, \quad \text{whenever } \sigma \geq 1 \text{ and } |t| \geq 1.$$

We now consider two separate cases, depending on whether the inequality $\sigma - 1 \geq A|t|^{-5\epsilon}$ holds, for some appropriate constant A (whose value we choose later).

If this inequality does hold, then (3) immediately provides

$$|\zeta(\sigma + it)| \geq A'|t|^{-4\epsilon},$$

and it suffices to replace 4ϵ by ϵ to conclude the proof of the desired estimate, in this case.

If, however, $\sigma - 1 < A|t|^{-5\epsilon}$, then we first select $\sigma' > \sigma$ with $\sigma' - 1 = A|t|^{-5\epsilon}$. The triangle inequality then implies

$$|\zeta(\sigma + it)| \geq |\zeta(\sigma' + it)| - |\zeta(\sigma' + it) - \zeta(\sigma + it)|,$$

and an application of the mean value theorem, together with the estimates for the derivative of ζ obtained in the previous chapter, give

$$|\zeta(\sigma' + it) - \zeta(\sigma + it)| \leq c''|\sigma' - \sigma|\,|t|^{\epsilon} \leq c''|\sigma' - 1|\,|t|^{\epsilon}.$$

These observations, together with an application of (3) where we set $\sigma = \sigma'$, show that

$$|\zeta(\sigma + it)| \geq c'(\sigma' - 1)^{3/4}|t|^{-\epsilon/4} - c''(\sigma' - 1)|t|^{\epsilon}.$$

Now choose $A = (c'/(2c''))^4$, and recall that $\sigma' - 1 = A|t|^{-5\epsilon}$. This gives precisely

$$c'(\sigma' - 1)^{3/4}|t|^{-\epsilon/4} = 2c''(\sigma' - 1)|t|^{\epsilon},$$

and therefore

$$|\zeta(\sigma + it)| \geq A''|t|^{-4\epsilon}.$$

On replacing 4ϵ by ϵ, the desired inequality is established, and the proof of the proposition is complete.

2 Reduction to the functions ψ and ψ_1

In his study of primes, Tchebychev introduced an auxiliary function whose behavior is to a large extent equivalent to the asymptotic distribution of primes, but which is easier to manipulate than $\pi(x)$. **Tchebychev's ψ-function** is defined by

$$\psi(x) = \sum_{p^m \leq x} \log p.$$

The sum is taken over those integers of the form p^m that are less than or equal to x. Here p is a prime number and m is a positive integer. There are two other formulations of ψ that we shall need. First, if we define

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^m \text{ for some prime } p \text{ and some } m \geq 1, \\ 0 & \text{otherwise,} \end{cases}$$

then it is clear that

$$\psi(x) = \sum_{1 \leq n \leq x} \Lambda(n).$$

Also, it is immediate that

$$\psi(x) = \sum_{p \leq x} \left[\frac{\log x}{\log p} \right] \log p$$

where $[u]$ denotes the greatest integer $\leq u$, and the sum is taken over the primes less than x. This formula follows from the fact that if $p^m \leq x$, then $m \leq \log x / \log p$.

The fact that $\psi(x)$ contains enough information about $\pi(x)$ to prove our theorem is given a precise meaning in the statement of the next proposition. In particular, this reduces the prime number theorem to a corresponding asymptotic statement about ψ.

Proposition 2.1 *If* $\psi(x) \sim x$ *as* $x \to \infty$, *then* $\pi(x) \sim x/\log x$ *as* $x \to \infty$.

Proof. The argument here is elementary. By definition, it suffices to prove the following two inequalities:

$$(4) \qquad 1 \leq \liminf_{x \to \infty} \pi(x) \frac{\log x}{x} \quad \text{and} \quad \limsup_{x \to \infty} \pi(x) \frac{\log x}{x} \leq 1.$$

To do so, first note that crude estimates give

$$\psi(x) = \sum_{p \leq x} \left[\frac{\log x}{\log p} \right] \log p \leq \sum_{p \leq x} \frac{\log x}{\log p} \log p = \pi(x) \log x,$$

and dividing through by x yields

$$\frac{\psi(x)}{x} \leq \frac{\pi(x) \log x}{x}.$$

The asymptotic condition $\psi(x) \sim x$ implies the first inequality in (4). The proof of the second inequality is a little trickier. Fix $0 < \alpha < 1$, and note that

$$\psi(x) \geq \sum_{p \leq x} \log p \geq \sum_{x^\alpha < p \leq x} \log p \geq (\pi(x) - \pi(x^\alpha)) \log x^\alpha,$$

and therefore

$$\psi(x) + \alpha\pi(x^\alpha) \log x \geq \alpha\pi(x) \log x.$$

Dividing by x, noting that $\pi(x^\alpha) \leq x^\alpha$, $\alpha < 1$, and $\psi(x) \sim x$, gives

$$1 \geq \alpha \limsup_{x \to \infty} \pi(x) \frac{\log x}{x}.$$

Since $\alpha < 1$ was arbitrary, the proof is complete.

Remark. The converse of the proposition is also true: if $\pi(x) \sim x/\log x$ then $\psi(x) \sim x$. Since we shall not need this result, we leave the proof to the interested reader.

In fact, it will be more convenient to work with a close cousin of the ψ function. Define the function ψ_1 by

$$\psi_1(x) = \int_1^x \psi(u)\, du.$$

In the previous proposition we reduced the prime number theorem to the asymptotics of $\psi(x)$ as x tends to infinity. Next, we show that this follows from the asymptotics of ψ_1.

Proposition 2.2 *If $\psi_1(x) \sim x^2/2$ as $x \to \infty$, then $\psi(x) \sim x$ as $x \to \infty$, and therefore $\pi(x) \sim x/\log x$ as $x \to \infty$.*

Proof. By Proposition 2.1, it suffices to prove that $\psi(x) \sim x$ as $x \to \infty$. This will follow quite easily from the fact that if $\alpha < 1 < \beta$, then

$$\frac{1}{(1-\alpha)x} \int_{\alpha x}^x \psi(u)\, du \leq \psi(x) \leq \frac{1}{(\beta-1)x} \int_x^{\beta x} \psi(u)\, du.$$

The proof of this double inequality is immediate and relies simply on the fact that ψ is increasing. As a consequence, we find, for example, that

$$\psi(x) \leq \frac{1}{(\beta-1)x}[\psi_1(\beta x) - \psi_1(x)],$$

and therefore

$$\frac{\psi(x)}{x} \leq \frac{1}{(\beta-1)}\left[\frac{\psi_1(\beta x)}{(\beta x)^2}\beta^2 - \frac{\psi_1(x)}{x^2}\right].$$

In turn this implies

$$\limsup_{x\to\infty} \frac{\psi(x)}{x} \leq \frac{1}{\beta-1}\left[\frac{1}{2}\beta^2 - \frac{1}{2}\right] = \frac{1}{2}(\beta+1).$$

Since this result is true for all $\beta > 1$, we have proved that $\limsup_{x\to\infty} \psi(x)/x \leq 1$. A similar argument with $\alpha < 1$, then shows that $\liminf_{x\to\infty} \psi(x)/x \geq 1$, and the proof of the proposition is complete.

It is now time to relate ψ_1 (and therefore also ψ) and ζ. We proved in Lemma 1.3 that for $\mathrm{Re}(s) > 1$

$$\log \zeta(s) = \sum_{m,p} \frac{p^{-ms}}{m}.$$

Differentiating this expression gives

$$\frac{\zeta'(s)}{\zeta(s)} = -\sum_{m,p} (\log p) p^{-ms} = -\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}.$$

We record this formula for $\mathrm{Re}(s) > 1$ as

(5) $$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}.$$

The asymptotic behavior $\psi_1(x) \sim x^2/2$ will be a consequence via (5) of the relationship between ψ_1 and ζ, which is expressed by the following noteworthy integral formula.

Proposition 2.3 *For all $c > 1$*

(6) $$\psi_1(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)}\left(-\frac{\zeta'(s)}{\zeta(s)}\right) ds.$$

To make the proof of this formula clear, we isolate the necessary contour integrals in a lemma.

Lemma 2.4 *If $c > 0$, then*

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{a^s}{s(s+1)}\, ds = \begin{cases} 0 & \text{if } 0 < a \leq 1, \\ 1 - 1/a & \text{if } 1 \leq a. \end{cases}$$

Here, the integral is over the vertical line $\mathrm{Re}(s) = c$.

Proof. First note that since $|a^s| = a^c$, the integral converges. We suppose first that $1 \leq a$, and write $a = e^\beta$ with $\beta = \log a \geq 0$. Let

$$f(s) = \frac{a^s}{s(s+1)} = \frac{e^{s\beta}}{s(s+1)}.$$

Then $\mathrm{res}_{s=0} f = 1$ and $\mathrm{res}_{s=-1} f = -1/a$. For $T > 0$, consider the path $\Gamma(T)$ shown on Figure 1.

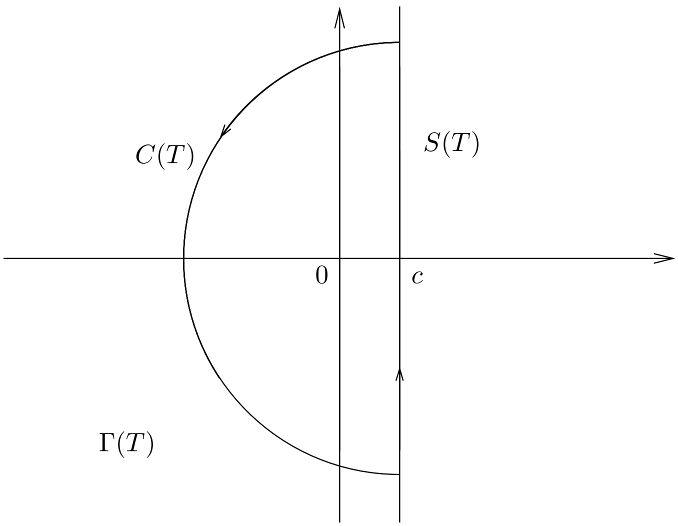

Figure 1. The contour in the proof of Lemma 2.4 when $a \geq 1$

The path $\Gamma(T)$ consists of the vertical segment $S(T)$ from $c - iT$ to $c + iT$, and of the half-circle $C(T)$ centered at c of radius T, lying to the left of the vertical segment. We equip $\Gamma(T)$ with the positive (counterclockwise) orientation, and note that we are dealing with a toy contour. If we choose T so large that 0 and -1 are contained in the interior of $\Gamma(T)$, then by the residue formula

$$\frac{1}{2\pi i}\int_{\Gamma(T)} f(s)\, ds = 1 - 1/a.$$

Since

$$\int_{\Gamma(T)} f(s)\, ds = \int_{S(T)} f(s)\, ds + \int_{C(T)} f(s)\, ds,$$

it suffices to prove that the integral over the half-circle goes to 0 as T tends to infinity. Note that if $s = \sigma + it \in C(T)$, then for all large T we have

$$|s(s+1)| \geq (1/2)T^2,$$

and since $\sigma \leq c$ we also have the estimate $|e^{\beta s}| \leq e^{\beta c}$. Therefore

$$\left| \int_{C(T)} f(s)\, ds \right| \leq \frac{C}{T^2} 2\pi T \to 0 \qquad \text{as } T \to \infty,$$

and the case when $a \geq 1$ is proved.

If $0 < a \leq 1$, consider an analogous contour but with the half-circle lying to the right of the line $\mathrm{Re}(s) = c$. Noting that there are no poles in the interior of that contour, we can give an argument similar to the one given above to show that the integral over the half-circle also goes to 0 as T tends to infinity.

We are now ready to prove Proposition 2.3. First, observe that

$$\psi(u) = \sum_{n=1}^{\infty} \Lambda(n) f_n(u),$$

where $f_n(u) = 1$ if $n \leq u$ and $f_n(u) = 0$ otherwise. Therefore,

$$\begin{aligned} \psi_1(x) &= \int_0^x \psi(u)\, du \\ &= \sum_{n=1}^{\infty} \int_0^x \Lambda(n) f_n(u)\, du \\ &= \sum_{n \leq x} \Lambda(n) \int_n^x du, \end{aligned}$$

and hence

$$\psi_1(x) = \sum_{n \leq x} \Lambda(n)(x - n).$$

This fact, together with equation (5) and an application of Lemma 2.4 (with $a = x/n$), gives

$$\begin{aligned}\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)}\left(-\frac{\zeta'(s)}{\zeta(s)}\right) ds &= x\sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{(x/n)^s}{s(s+1)}\,ds \\ &= x\sum_{n\le x}\Lambda(n)\left(1-\frac{n}{x}\right) \\ &= \psi_1(x),\end{aligned}$$

as was to be shown.

2.1 Proof of the asymptotics for ψ_1

In this section, we will show that

$$\psi_1(x) \sim x^2/2 \quad \text{as } x \to \infty,$$

and as a consequence, we will have proved the prime number theorem.

The key ingredients in the argument are:

- the formula in Proposition 2.3 connecting ψ_1 to ζ, namely

  $$\psi_1(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{x^{s+1}}{s(s+1)}\left(-\frac{\zeta'(s)}{\zeta(s)}\right) ds$$

  for $c > 1$.

- the non-vanishing of the zeta function on $\mathrm{Re}(s) = 1$,

  $$\zeta(1+it) \neq 0 \quad \text{for all } t \in \mathbb{R},$$

  and the estimates for ζ near that line given in Proposition 2.7 of Chapter 6 together with Proposition 1.6 of this chapter.

Let us now discuss our strategy in more detail. In the integral above for $\psi_1(x)$ we want to change the line of integration $\mathrm{Re}(s) = c$ with $c > 1$, to $\mathrm{Re}(s) = 1$. If we could achieve that, the size of the factor x^{s+1} in the integrand would then be of order x^2 (which is close to what we want) instead of x^{c+1}, $c > 1$, which is much too large. However, there would still be two issues that must be dealt with. The first is the pole of $\zeta(s)$ at $s = 1$; it turns out that when it is taken into account, its contribution is exactly the main term $x^2/2$ of the asymptotic of $\psi_1(x)$. Second, what remains must be shown to be essentially smaller than this term, and so